Grasp-TAGS: Task-Aware Grasp Selection via Critic Distillation for Prehensile Loco-Manipulation

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Introduction

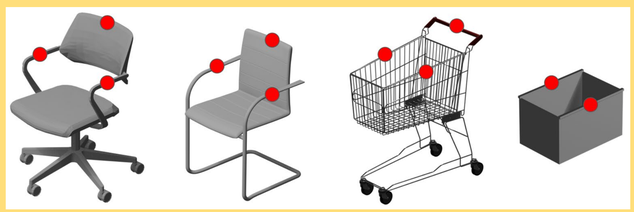
Different initial grasps can lead to substantially different outcomes in large-object loco-manipulation. **Grasp-TAGS** trains one PPO [1] policy across multiple initial grasp configurations and uses the learned critic to estimate their associated downstream task utility. The critic is distilled into a lightweight grasp utility network using only pre-interaction grasp and goal information. At inference time, the grasp with the highest predicted utility is selected for execution.

Pipeline

- **Multi-grasp pipeline training:** Train one PPO loco-manipulation policy across multiple feasible initial grasp configurations, where the PPO critic can capture the expected downstream utility of each grasp.
- **Critic distillation:** Distill the critic into a lightweight grasp utility network using only grasp and goal information.
- **Grasp selection at inference:** Rank candidate grasps, select the highest-utility grasp, and execute the unified manipulation policy.

Dataset information

We evaluate on four synthetic household objects from Fab [2], assigned multiple predefined feasible grasp configurations



Preliminary Experiments

Grasp-TAGS consistently achieves the highest success rate across all evaluated objects, outperforming both fixed single-grasp policies and random grasp selection.

Reference:

[1] Schulman *et al.*, “Proximal Policy Optimization Algorithms,” *arXiv:1707.06347*, 2017.
[2] Epic Games, Inc., “Fab,” 2026, 3D asset marketplace.

Our method selects the best grasp with the highest predicted downstream utility before policy execution

Candidate grasp poses

- g_1: backrest
- g_2: right armrest
- g_3: left armrest



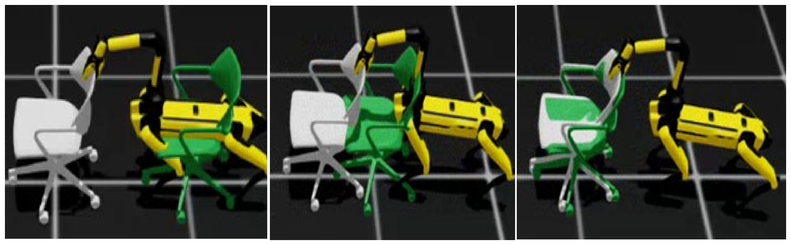
Critic-distilled Grasp Selector

Rank candidate grasps (by predicted downstream utility)

$g_1 > g_2 > g_3$

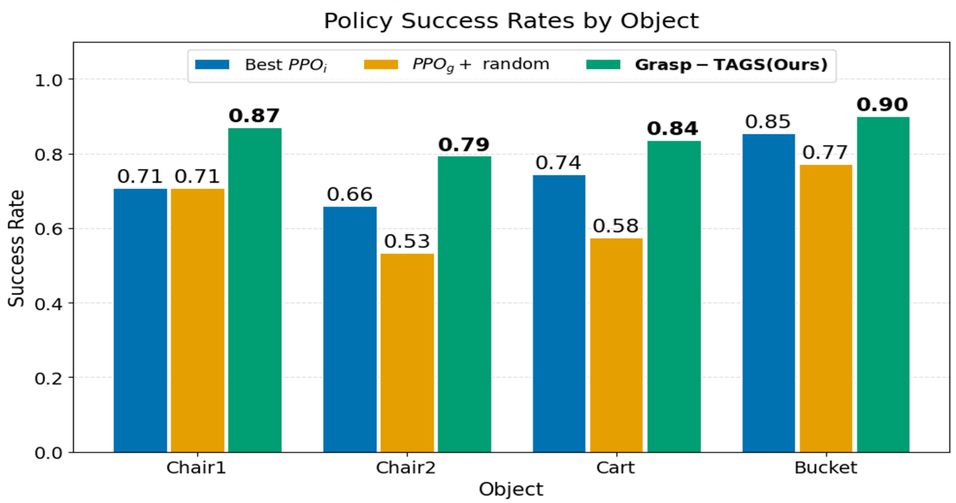
Select g_1 (best)

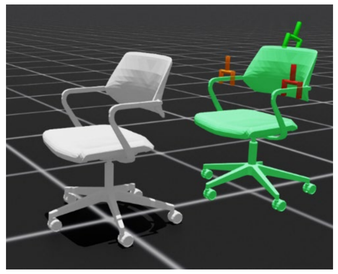
Execute (with grasp g_1)



Successful rearrangement ✓

Preliminary Results





Grasp candidates ranked on the target object

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